CHAPTER 9

IPv4 Addressing, CIDR & Subnetting

Classes · masks · network and broadcast address · hosts per block · CIDR · aggregation · VLSM

QUICK TRICK

Block size = $2^{(32-n)}$ for /n. Usable hosts = **block − 2**. Network address = IP **AND** mask. Broadcast = network + block − 1. A block must **start at a multiple of its own size**.

9.1 IPv4 address basics

● **LEVEL 1 · BEGINNER**

An IPv4 address has **32 bits**, written as four decimal numbers 0–255 (dotted decimal). It has a **network part** (prefix) and a **host part**. Example: 192.168.10.37 = 11000000.10101000.00001010.00100101.

Classful addressing (old, but still asked)

Class	First bits	First octet	Default mask	Hosts per network
A	0	1 – 126	/8 (255.0.0.0)	$2^{24} - 2$
B	10	128 – 191	/16 (255.255.0.0)	$2^{16} - 2$
C	110	192 – 223	/24 (255.255.255.0)	254
D	1110	224 – 239	multicast	–
E	1111	240 – 255	reserved	–

Special address	Meaning
0.0.0.0	This host / default route
127.0.0.0/8	Loopback (127.0.0.1)
255.255.255.255	Limited broadcast (this network)
10.0.0.0/8, 172.16.0.0/12 (172.16–172.31), 192.168.0.0/16	Private (not routed on the Internet)
169.254.0.0/16	Link-local (auto-configuration)

9.2 Masks, network and broadcast

● **LEVEL 2 · EXAM**

- **Mask** = n ones then 32 − n zeros. /27 = 255.255.255.224.
- **Network address** = IP AND mask (host bits all 0). **Broadcast** = host bits all 1. **Usable hosts** = $2^h - 2$ where h = host bits.
- First usable = network + 1. Last usable = broadcast − 1.

Prefix	Last mask octet	Block size	Usable hosts
/24	0	256	254
/25	128	128	126
/26	192	64	62
/27	224	32	30
/28	240	16	14
/29	248	8	6
/30	252	4	2 (point-to-point links)

Quick method (the "magic octet"). Find the octet where the mask is not 255 or 0, get block size = 256 − mask value, then network = largest multiple of the block size ≤ that octet. **Example:** 172.16.37.200/20 → mask 255.255.240.0, block = 16 in the 3rd octet, 37 → 32. Network **172.16.32.0**, broadcast **172.16.47.255**.

9.3 CIDR (classless)

● LEVEL 2 · EXAM

CIDR drops the classes: any prefix length n (a.b.c.d/n). It removes waste (a company needing 2000 addresses gets a /21, not a whole class B) and lets routers **aggregate** routes.

Aggregation (supernetting). Several contiguous blocks can be combined into one **if** (1) their count is a power of 2, (2) they are contiguous, (3) the first starts at a multiple of the total size. **Yes:** 192.168.0.0/24 … 192.168.3.0/24 → **192.168.0.0/22**. **No:** 192.168.1.0/24 + 192.168.2.0/24 (starts at 1, not a multiple of 2). 10.1.2.0/24 + 10.1.3.0/24 → **10.1.2.0/23**.

9.4 Subnetting and VLSM

● LEVEL 3 · GATE

- **Fixed-length subnetting:** borrow s host bits → 2^s equal subnets of 2^{h-s} addresses.
- **VLSM (variable-length):** give each LAN the **smallest block that fits** (hosts + 2). **Allocate the largest first**, from the lowest address, so blocks stay aligned.

Example. Divide 192.168.10.0/24 for LANs of 100, 50, 20, 10 hosts:

LAN	Hosts + 2	Block	Prefix	Range
1	102	128	/25	192.168.10.0 – .127
2	52	64	/26	192.168.10.128 – .191
3	22	32	/27	192.168.10.192 – .223
4	12	16	/28	192.168.10.224 – .239
free		16	/28	192.168.10.240 – .255

9.5 Common mistakes

COMMON MISTAKE

- Forgetting the −2 (network and broadcast addresses) for hosts, or forgetting to add 2 when sizing a block.
- Starting a block at an address that is not a multiple of its size.
- Treating 172.32.x.x as private (the private range is 172.16–172.31).
- Using classful rules when a CIDR prefix is given.

QUICK TRICK

Memory line: "Biggest block first, aligned to its own size."

GATE QUICK CHECK — Chapter 9

```
/n -> mask n ones | block = 2^(32-n) | hosts = block-2 | net = IP AND mask
/25=128 /26=192 /27=224 /28=240 /29=248 /30=252  (last octet)   blocks 128 64 32 16 8 4
Class A 1-126 | B 128-191 | C 192-223 | D 224-239 | E 240-255
Aggregate: power-of-2 count, contiguous, first block aligned to total size
```

Practice questions

● LEVEL 1 · BEGINNER

Q1. Usable hosts in a /26.

Q2. Class of 130.5.6.7.

Q3. Mask for /27 in dotted decimal.

Q4. Is 172.32.0.1 a private address?

● LEVEL 2 · EXAM

Q5. Network and broadcast address of 172.16.37.200/20.

Q6. Host range of 192.168.1.130/26.

Q7. How many addresses and usable hosts in a /22?

Q8. Mask 255.255.255.240 on a class C network: number of subnets and hosts each?

Q9. Can 10.1.2.0/24 and 10.1.3.0/24 be aggregated?

● **LEVEL 3 · GATE**

Q10. Smallest CIDR block for 2000 hosts.

Q11. An organisation has 200.10.0.0/22. Allocate to departments of 500, 200 and 100 hosts.

Q12. Which of 192.168.1.0/24 + 192.168.2.0/24 or 192.168.2.0/24 + 192.168.3.0/24 can be aggregated?

Q13. How many /30 subnets fit in a /24, and how many hosts in total can they hold?

Answers

Q	Solution
1	$2^6 - 2 =$ **62**.
2	First octet 130 is in 128–191: **Class B**.
3	**255.255.255.224**.
4	**No.** Private is 172.16.0.0 – 172.31.255.255.
5	Network **172.16.32.0**, broadcast **172.16.47.255**.
6	Block 64: network 192.168.1.128, hosts **.129 – .190**, broadcast .191.
7	$2^{10} =$ **1024** addresses, **1022** hosts.
8	4 bits borrowed → **16 subnets**, $2^4 - 2 =$ **14 hosts** each.
9	**Yes** → 10.1.2.0/23 (2 is a multiple of 2).
10	$2000 + 2 = 2002 \le 2^{11} = 2048$, so **/21**.
11	500 → /23 (512): **200.10.0.0/23**. 200 → /24: **200.10.2.0/24**. 100 → /25: **200.10.3.0/25**. Left: 200.10.3.128/25.
12	Only the second: **192.168.2.0/23**. (192.168.1.0 is not a multiple of the /23 size.)
13	$2^6 =$ **64** subnets, each 2 usable → **128** hosts in all.

Next: Chapter 10 — IPv4 Header, Fragmentation and Forwarding (longest-prefix match).